Week 8 Assignment Report-
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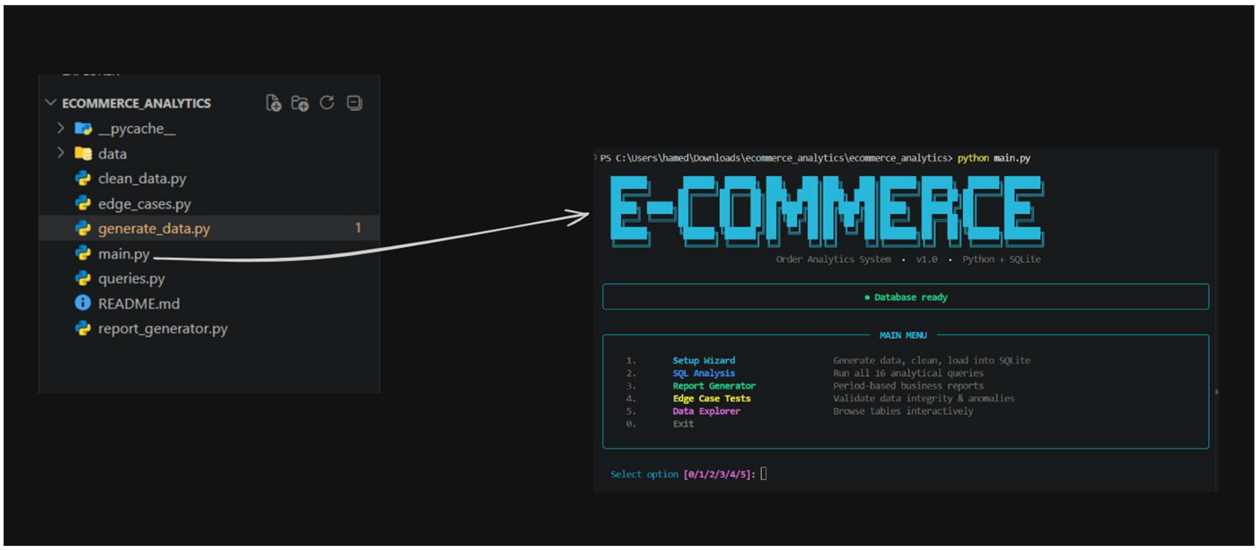
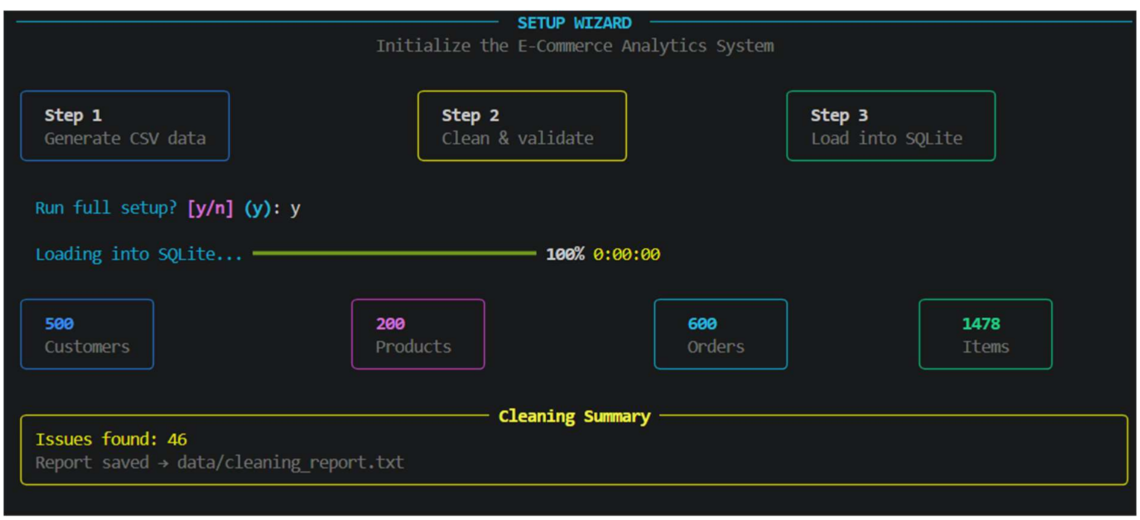
This is a small end-to-end e-commerce analytics project built in Python. It demonstrates a complete data workflow: generate messy sample data, clean it, load it into SQLite, run SQL analysis, and generate business reports.



A terminal-based UI in main.py with a guided setup flow made using rich python module.

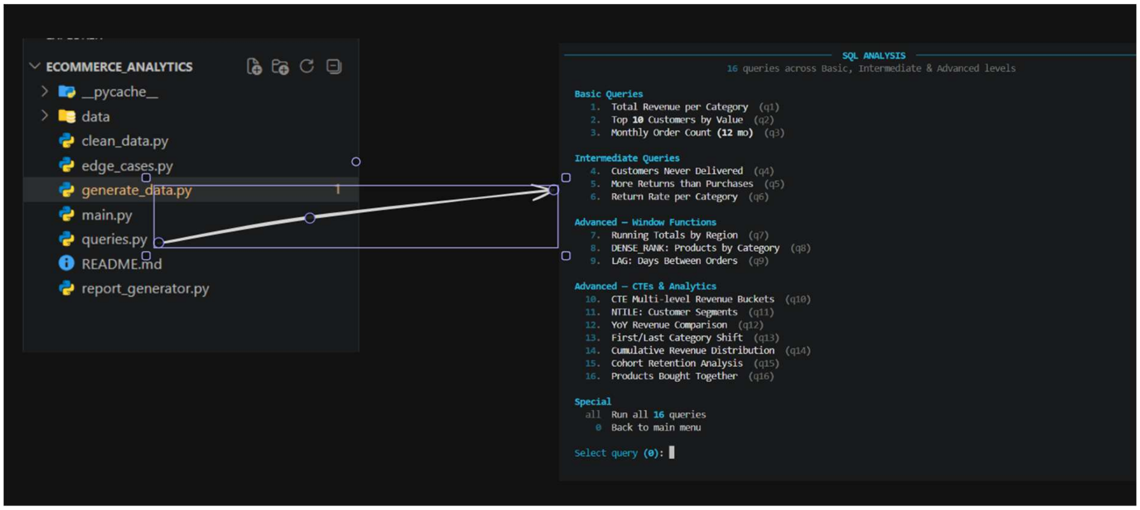


A setup wizard that performs three steps
1.Generate syntetic csv data using the generate_data.py on purpose with some problems in it, so the project can show how messy real data looks.
2.Clean and validate data , this fixes the messy data by cleaning dates, filling missing customer IDs, validating emails, and removing broken order items using the clean_data.py

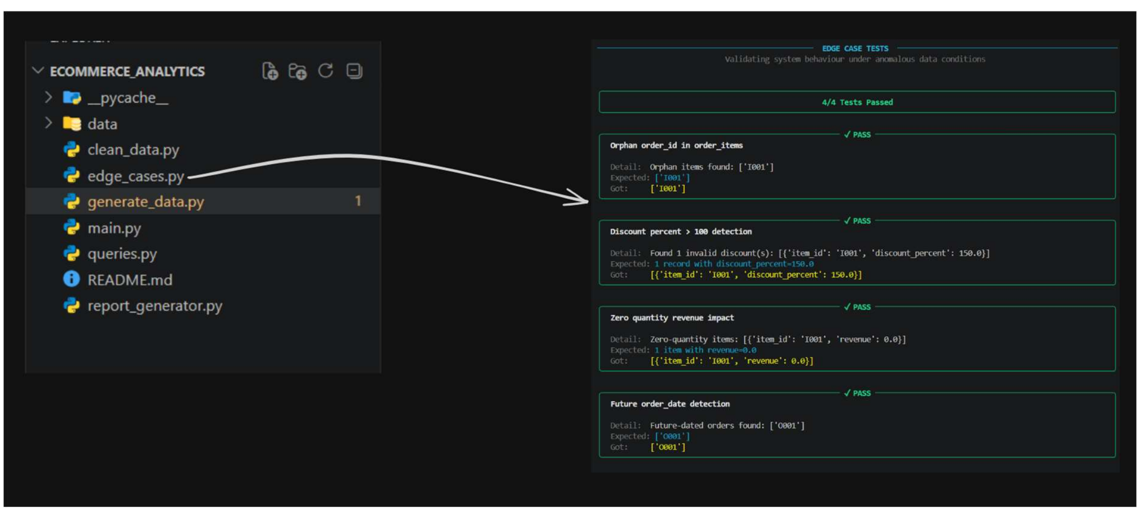
3.Finally the last step is loading data in SQLite:
So the flow is basically:
CSV files → read them → map them to table columns → insert into SQLite tables.

```
View issue log? [y/n] (n): y

#      Issue

1      NULL customer_id for order O00029
2      NULL customer_id for order O00055
3      NULL customer_id for order O00118
4      NULL customer_id for order O00124
5      NULL customer_id for order O00133
6      NULL customer_id for order O00205
7      NULL customer_id for order O00222
8      NULL customer_id for order O00232
9      NULL customer_id for order O00234
10     NULL customer_id for order O00239
11     NULL customer_id for order O00278
12     NULL customer_id for order O00365
13     NULL customer_id for order O00381
14     NULL customer_id for order O00398
15     NULL customer_id for order O00414
16     NULL customer_id for order O00433
17     NULL customer_id for order O00520
18     NULL customer_id for order O00526
19     NULL customer_id for order O00530
20     NULL customer_id for order O00560
21     NULL customer_id for order O00575
22     NULL customer_id for order O00579
23     NULL customer_id for order O00586
24     Normalized name 'Deluxe  Camera  970' -> 'Deluxe Camera 970' for P
25     Normalized name 'Ultra  Frame  888' -> 'Ultra Frame 888' for P0030
26     Normalized name 'elite collection 225' -> 'Elite Collection 225' f
27     Normalized name 'ADVANCED JEANS 705' -> 'Advanced Jeans 705' for P
28     Normalized name 'Smart Masterclass 562' -> 'Smart Masterclass 562'
29     Normalized name 'ULTRA TABLET 260' -> 'Ultra Tablet 260' for P0090
30     Normalized name 'Smart Volume 406' -> 'Smart Volume 406' for P0099
...    and 16 more (see data/cleaning_report.txt)

✓ Setup complete! System ready for analysis.
```

The raw ecommerce data was cleaned by standardizing dates, filling missing customer IDs, validating email addresses, normalizing product names, and removing invalid order-item records so the dataset was ready for analysis



Queries.py converts cleaned data into actionable business insights.
It performs two primary tasks:
(1) creating a SQLite database and loading cleaned CSV files into tables for customers, products, orders, and order_items; and
(2) executing SQL queries to address business questions.
Key example queries and analyses include:
•Revenue by product category
•Identifying top customers by spending
•Monthly order counts
•Customers who never received orders
•Products with more returns than purchases
•Return rates by category
•Running revenue totals and rankings
•Trend and comparison analyses such as lag/lead and year-over-year comparisons



Edge_cases.py It handles things like:

- orphaned order items that point to a non-existent order
- discount percentages above 100%, which are invalid
- zero-quantity items that may distort revenue
- orders with future dates, which could affect time-based analysis

```
DATA EXPLORER
Browse your raw database tables

1.  customers
2.  orders
3.  order_items
4.  products
0.  Back

Select table (0): 1
Rows to display (20): 20
```

CUSTOMERS

customer_id	customer_name	email	customer_type	registration_date
C0001	James Gupta	james.gupta26@hotmail.com	VIP	2020-08-04
C0002	Vihaan Davis	vihaan.davis90@rediffmail.com	REGULAR	2020-02-16
C0003	Reyansh Shah	reyansh.shah518@rediffmail.com	VIP	2020-11-04
C0004	James Davis	james.davis430@yahoo.com	REGULAR	2021-03-08
C0005	Emma Kumar	emma.kumar715@outlook.com	REGULAR	2020-08-29
C0006	Emma Smith	emma.smith105@gmail.com	PREMIUM	2021-07-11
C0007	Megha Joshi	megha.joshi827@gmail.com	REGULAR	2022-04-13
C0008	Charlotte Williams	charlotte.williams81@rediffmail.com	VIP	2022-09-03
C0009	Sofia Johnson	sofia.johnson592@yahoo.com	VIP	2020-03-12
C0010	Ayaan Nair	ayaan.nair82@yahoo.com	PREMIUM	2021-08-13
C0011	Riya Johnson	riya.johnson167@hotmail.com	PREMIUM	2022-11-07
C0012	Harry Patel	harry.patel624@yahoo.com	REGULAR	2021-01-15
C0013	Riya Williams	riya.williams277@rediffmail.com	REGULAR	2021-05-18
C0014	Ayaan Verma	ayaan.verma825@hotmail.com	REGULAR	2020-04-12
C0015	Charlotte Wilson	charlotte.wilson898@hotmail.com	PREMIUM	2022-02-14
C0016	Sofia Jones	sofia.jones147@hotmail.com	VIP	2023-03-03
C0017	Shruti Joshi	shruti.joshi765@rediffmail.com	PREMIUM	2022-06-25
C0018	Ananya Shah	ananya.shah142@rediffmail.com	REGULAR	2023-03-20
C0019	Isabella Gupta	isabella.gupta157@yahoo.com	VIP	2021-10-17
C0020	Aditya Williams	aditya.williams391@rediffmail.com	PREMIUM	2022-04-01

```
✓ 20 row(s) returned
i Table has 500 total rows
```

The data explorer part in main.py is the section it:

- runs or displays the analytics queries from queries.py
- presents the results in a clear terminal view using tables and panels
- helps you explore business questions such as sales, customers, returns, and trends